Chunxiao Jing

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Agricultural and Applied Economics, University of Wisconsin-Madison

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Education

- **University of Wisconsin-Madison** *August, 2021-May, 2027(expected)*
  Ph.D., Agricultural and Applied Economics Madison, US
  Committee: Jeremy Foltz (Chair), Gisella Kagy, Vladimir Gimpelson
- **University of California, Los Angeles** *June, 2019*
  M.A., Applied Economics Los Angeles, US
- **Nankai University** *June, 2018*
  B.A., Economics Tianjin, China

Fields of Interest

Primary Fields: Development Economics, Applied Microeconomics
Secondary Fields: Labor Economics, Structural Change, Environmental Economics

Job Market Paper

Weather Shocks, Labor Reallocation, and Structural Transformation: Micro Evidence from Africa

Abstract. Structural transformation, the movement of labor from agriculture to more productive non-agricultural activities, is a central driver of economic development. Classical theories emphasize different stages of this process, including labor release from agriculture (Lewis, 1954), labor reallocation under transition frictions, and labor absorption into non-agricultural employment (Harris and Todaro, 1970). This paper develops a structural framework that unifies these perspectives by decomposing structural transformation into three sequential stages: labor release, labor reallocation, and labor absorption. Using household panel data from the Living Standards Measurement Study (LSMS) for Niger, Malawi, and Ethiopia, combined with high-resolution climate and geospatial data, I first provide reduced-form evidence on how weather-induced agricultural productivity shocks using Random Forest affect each stage of the transformation process. I then estimate a structural model using the simulated method of moments and conduct counterfactual policy analysis. The results reveal substantial heterogeneity in the mechanisms governing structural transformation across countries. Subsistence constraints are the primary barrier in Niger and Malawi, whereas labor reallocation frictions and limited labor absorption constrain structural transformation in Ethiopia. These findings suggest that no single policy can effectively promote structural transformation across countries. By identifying the binding constraint at each stage, the proposed framework provides a unified approach for understanding labor allocation and designing country-specific development policies.

Working Papers

Can the Service Sector Lead Structural Transformation in Africa? Evidence from Côte d'Ivoire [Link]
with Jeremy Foltz
In this paper, we examine whether expansion of the service sector can become a driver of structural transformation in Côte d'Ivoire by analyzing labor demand and sectoral productivity. Using proxy-variable estimates of total factor productivity, we find that service-sector productivity is 2.46 times higher than manufacturing productivity, and that high-productivity service firms hire more workers overall, particularly unskilled workers. The results suggest that the service sector can lead structural transformation and GDP growth in Africa.

Work in Progress

Gender Constraints and Labor Allocation during Structural Transformation

In this paper, I explore whether gender differences in sectoral labor allocation reflect preferences or gender-specific barriers to non-agricultural employment. Using LSMS panel data from Niger, Nigeria, Malawi, and Ethiopia combined with weather-induced agricultural productivity shocks, the project estimates how men and women differ in their sectoral transitions.

Climate Change and Conflict: Causal Evidence with New Machine-Learning Methods
with Jeremy Foltz

In this paper, we study how climate shocks affect conflict through direct and spatial spillover channels across ecological systems in Africa. The project combines geospatial panel data with Double/Debiased Machine Learning and an exposure-mapping framework that accounts for neighboring climate shocks. We find that rainfall shocks generate conflict primarily through spillover channels, whereas temperature shocks mainly affect directly exposed locations. Our results demonstrate that accounting for spatial spillovers and ecological heterogeneity substantially improves understanding of climate-conflict dynamics and helps explain why previous studies reach conflicting conclusions.

Research Experience

University of Wisconsin–Madison — Madison, WI
Research Assistant to Prof. Jeremy Foltz — Fall 2022 – Present

University of California, Los Angeles — Los Angeles, CA
Research Assistant, MAE Research Lab — Spring 2019

Teaching Experience

- **The Growth and Development of Nations in the Global Economy (AAE 374)** — *Teaching Assistant*
  Taught by Professor Vladimir Gimpelson — Spring, 2024
- **The Growth and Development of Nations in the Global Economy (AAE 374)** — *Teaching Assistant*
  Taught by Professor Jeremy Foltz — Spring, 2023

Conferences and Presentations

* indicates upcomming

2026 MEA Annual Meeting, AAEA Annual Meeting*
2025 ASSA Annual Meeting, AAE Development Workshop
2024 CSAE Conference, CES China Annual Conference, AAEA Annual Meeting
2023 AAE Development Workshop

FUNDINGS AND GRANTS

AAE Research Travel Grant, 2026-2027, $1,000
Graduate Student Research Grants, Research Travel Funds, 2026-2027, $1,500
STEG Small Research Grant, 2026-2027, £5,000
STEG PhD Research Grant, 2026-2027, £10,000
UW-Madison PhD Fellowship, 2025-2026, $35,636
AAE Conference Travel Grant, 2024-2025, $1,000
AAE Conference Travel Grant, 2023-2024, $1,000
UW-Madison PhD Fellowship, 2021-2022, $30,545

Professional Service

- **Mentor Service**
  Mentor, The Graduate Applications International Network (GAIN) Program — 2022-2023 and 2024-2025
- **Department Service**
  The Taylor-Hibbard Club President, University of Wisconsin-Madison — 2023-2024

Language and Computational Skills

Languages: Chinese (Native), English (Fluent)
Computer Skills: Python, Stata, R, LaTeX, QGIS, SPSS
Methodological Expertise: Causal Inference, Machine Learning, Structural Modeling

References

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